

Learning When Black-Hole Hair Is Observable from Synthetic Ringdown Waves

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Abstract—We ask when a synthetic black-hole ringdown contains enough information to recover the Kiselev parameters (k, w_q) , and when the inverse map is too ill-conditioned for finite-precision inference. To prevent repeated mode realizations of one spacetime from leaking across data partitions, we use grouped physical cross-validation and fit waveform PCA only on training folds. A waveform-only random forest reaches $R^2(k) = 0.8960$ and $R^2(w_q) = 0.9587$; the original synthetic-geodesic proxy experiment raises these scores to 0.9793 and 0.9891 with histogram gradient boosting. More importantly, an all-feature random-forest classifier reproduces a Fisher/Jacobian observability boundary with macro F1 = 0.9928, using $\sigma_{w_q} > 0.3$ at one-percent relative observable precision as the weak-identifiability criterion. In a companion scalar/proxy identifiability experiment, adding synthetic geodesic features reduces mean absolute w_q error from 0.0225 to 0.0143 and improves local conditioning near $k = 0$. We subsequently replace those proxies with validated timelike-emitter and three-dimensional null-geodesic shooting. On a 121-point physical Kiselev grid, adding photon geometry to ringdown improves the median minimum singular value by $4.54\times$ and median conditioning by $2.28\times$ relative to ringdown alone. This is a controlled physical-identifiability benchmark, not an observational constraint. Final inverse validation confirms lower grouped w_q error for the combined set in both tree families, but directional extrapolation is substantially harder and 81-to-161-phase feature substitution changes selected tree predictions materially. The result therefore establishes physical complementarity while identifying a remaining numerical-robustness condition for final inverse-model claims.

Index Terms—scientific machine learning, inverse problems, black-hole ringdown, identifiability, grouped validation, Fisher information

I. Introduction

This paper studies a specific inverse problem: recovering the Kiselev parameters (k, w_q) from the frequency and damping morphology of synthetic ringdown curves. In the eikonal limit, quasinormal frequencies are related to the orbital frequency and instability exponent of an unstable circular null geodesic [1], [2]. The relation gives a transparent simulator in which machine-learning behavior can be compared directly with the local derivatives of the observable map.

In this setting, a high R^2 can mean either genuine recovery of Kiselev parameters or interpolation across repeated mode realizations of the same spacetime. Each (k, w_q) point produces several curves through different angular indices ℓ and overtones n ; a row-wise split can

therefore place nearly identical physical systems in training and test sets. Even after that leakage is removed, a model may interpolate accurately in most of the grid while becoming unreliable where the inverse map is locally ill-conditioned. Our central question is consequently: when does a synthetic ringdown contain enough information to infer Kiselev hair parameters, and when are finite-precision errors amplified too strongly?

We address these issues in a synthetic Kiselev black-hole study. The Kiselev solution is a static, spherically symmetric geometry containing an amplitude k and state parameter w_q [3]. Our simulator uses leading-order analytic shifts rather than a dedicated perturbative solver. We generate damped curves, enforce grouped physical validation, fit PCA within training folds, and compare several inverse models. We then change the target: instead of predicting only (k, w_q) , a classifier predicts whether w_q is practically identifiable under a stated precision model. A numerical Fisher/Jacobian calculation provides the scientific label and an analytic interpretation of failures.

We address that question through the original inverse-learning study and a subsequent physical-validation program:

- grouped physical validation for waveform-to-hair inference, with PCA fitted inside each training fold;
- regression from synthetic leading-eikonal curves to (k, w_q) ;
- Fisher/Jacobian labels for weakly identifiable regions;
- comparison of waveform-only, scalar, and synthetic geodesic-proxy representations; and
- joint interpretation of ML errors and local singular-value diagnostics;
- Hamiltonian timelike and null shooting with physical phase beginning at apocentre, $\phi = \pi$;
- Schwarzschild, small- k , and fixed- k w_q -sensitivity validation; and
- two-dimensional physical Jacobian and observable-complementarity maps.

The central result is not the regression score by itself. The more useful outcome is that a learned classifier reproduces the Fisher/Jacobian observability boundary, separating regions where the waveform-to-hair map is informative from regions where finite-precision errors are

amplified.

II. Synthetic Leading-Eikonal Waveform Dataset

A. Static Kiselev simulator

We set the mass scale to $M = 1$ and sample a dense grid

$$k \in [-0.04, 0.04], \quad w_q \in [-1.4, 1.4]. \quad (1)$$

The underlying Kiselev metric family motivates the dependence on both parameters [3]; the present implementation uses analytic leading-order expressions for the photon-orbit radius shift Δr , orbital frequency Ω , and Lyapunov exponent λ . Non-finite or non-positive leading-order points are excluded. The broad w_q interval is used as a controlled benchmark of inverse-map geometry and is not intended to identify every sampled point with the standard quintessence regime.

For each retained physical point, the eikonal mapping is

$$\omega_R = \ell\Omega, \quad \gamma = (n + \tfrac{1}{2})\lambda, \quad (2)$$

and the synthetic time-domain curve is

$$\Psi(t) = \exp(-\gamma t) \cos(\omega_R t). \quad (3)$$

We use $\ell \in \{4, 5, 10, 20\}$, $n \in \{0, 1\}$, and 512 uniformly spaced time samples. The eight mode curves are averaged into one 512-sample multi-mode morphology per physical point for the waveform-to-hair experiment. This aggregation prevents repeated modes from being treated as independent target samples while retaining their combined morphology. The equal-weight average is a benchmark feature construction rather than an astrophysical multimode waveform with physically inferred amplitudes and phases.

B. Feature families

Five representations are evaluated: (A) the raw waveform; (B) waveform PCA; (C) waveform PCA with (Ω, λ) ; (D) waveform PCA with four synthetic geodesic proxies; and (E) waveform PCA with all current scalars and proxies. The current scalars are Ω , λ , ω_R , γ , and Δr .

The additional features are smooth proxies for impact parameter, screen coordinate, propagation-time delay, and a redshift curve. They exercise a replaceable multi-observable interface and test whether parameter dependence not reducible to (Ω, λ) can improve conditioning. They should not be interpreted as geodesic-shooting or ray-tracing predictions.

The later physical-shooting features are a distinct family, not a relabeling of these proxies. They are derived from integrated emitter and photon trajectories and include redshift, impact parameter, signed observer-tetrad sky coordinates, propagation delay, and arrival-time curves. The regression values below remain the historical proxy benchmark.

III. Leakage-Aware Inverse-Learning Protocol

A. Grouped physical validation

The grouping variable is the physical pair (k, w_q) ; neither ℓ nor n defines a new physical system. Grouped cross-validation therefore assigns all mode realizations of one physical point to exactly one fold. This is the central leakage control. For the zero-noise headline experiment, PCA is fitted on clean training curves. In the noise study, independent noise is added to training and test curves, and PCA is fitted only on the noisy training fold at each noise level. PCA never sees held-out test data. This follows the general train-transform-test discipline implemented with scikit-learn [4].

Regression candidates are Ridge, random forest, histogram gradient boosting, and a multilayer perceptron. Classification candidates are logistic regression, random forest, histogram gradient boosting, and a multilayer perceptron. Random forests provide a robust nonlinear baseline [5]; gradient boosting supplies a complementary additive-tree model [6]. Hyperparameters are deliberately modest because the scientific question concerns information content rather than model scaling.

Regression is measured using R^2 , mean absolute error (MAE), and root mean squared error for each target. Classification is measured with accuracy, macro F1, and a confusion matrix. Gaussian waveform noise at 0, 0.5, 1, 3, and 5 percent of the per-waveform standard deviation is injected before PCA or direct waveform input.

B. Practical-identifiability label

Let the local observable vector be

$$\mathbf{y}(k, w_q) = (\Omega, \lambda, \omega_R, \gamma, \Delta r)^\top. \quad (4)$$

Finite differences give the Jacobian

$$J = \frac{\partial \mathbf{y}}{\partial (k, w_q)}. \quad (5)$$

Under independent one-percent relative errors, with a small absolute floor for zero-valued shifts, we whiten the Jacobian as

$$\tilde{J} = C_y^{-1/2} J = U \text{diag}(s_i) V^\top. \quad (6)$$

Singular values below $\max(\epsilon_{\text{abs}}, \epsilon_{\text{rel}} s_{\text{max}})$, with $\epsilon_{\text{rel}} = 10^{-8}$ and $\epsilon_{\text{abs}} = 0$, define the numerical null space. This tolerance is supported by derivative convergence: five-point and three-point whitened Jacobians differ relatively by at most 9.31×10^{-9} at representative near-degenerate points, and scaling both finite-difference steps from 0.25 to 4 leaves all labels unchanged. If the explicit w_q direction overlaps the numerical null space, we set $\sigma_{w_q} = \infty$ and mark the parameter structurally or numerically non-identifiable. Otherwise the covariance is assembled from retained singular directions, $\Sigma_\theta = V_r \text{diag}(s_r^{-2}) V_r^\top$. This avoids the incorrect finite zero uncertainty that a bare pseudoinverse assigns to a discarded parameter direction.

TABLE I

Grouped waveform-to-hair results. Geodesic features are synthetic proxies, not physical ray-tracing outputs.

Input and best regressor	$R^2(k)$	$R^2(w_q)$
Waveform only, random forest	0.8960	0.9587
Waveform + proxies, HistGB	0.9793	0.9891

At exactly $k = 0$, the observables have no w_q dependence, the Jacobian has rank one, and $\sigma_{w_q} = \infty$. At small nonzero $|k|$, by contrast, the Jacobian can remain full rank while producing a large but finite uncertainty. We call this practical weak identifiability, labeling a point weak when

$$\sigma_{w_q} = \sqrt{(\Sigma_\theta)_{w_q w_q}} > 0.3. \quad (7)$$

This criterion labels 57.6% of proxy-valid physical points as weakly identifiable. It is a simulator-specific, precision-dependent phase diagram, not an observational posterior.

C. Reproducibility and audit trail

The experiment is implemented in the open research prototype black-hole-hair-ml [7]. Configuration files record the dense parameter ranges, mode indices, number of time samples, PCA dimension, model settings, and noise levels. Each run exports fold-level metrics, out-of-fold predictions, noise tables, paired PNG/PDF figures, and an automatically generated scope-aware report. The physical group identifier is stored explicitly with predictions, allowing a post-run check that no group crosses the train/test boundary. The PCA object is constructed inside the fold loop rather than fitted once on the full dataset. These implementation details are important because a numerically strong result would not be scientifically interpretable if preprocessing or repeated-mode leakage exposed test information during training.

The present workshop draft reports the completed default experiment rather than a hyperparameter optimization study. Model families were compared under the same grouped partitions, and the best named estimators are selected from that controlled comparison. The feature comparison reports the best mean grouped-CV score for each feature-set/target pair.

IV. Waveform-to-Hair Prediction Results

Table I summarizes the main grouped results. Waveform morphology alone is informative: the random forest reaches $R^2(k) = 0.8960$ and $R^2(w_q) = 0.9587$. Because Eq. (3) is fully determined by $(\Omega, \lambda, \ell, n)$, this performance measures how well the model inverts the frequency and damping morphology within the sampled simulator. It does not demonstrate additional physical information in the curve.

Adding proxy observables raises the jointly selected model scores to 0.9793 and 0.9891 with histogram gradient boosting. Figure 1 gives the complementary best-per-target comparison across representations. Within this

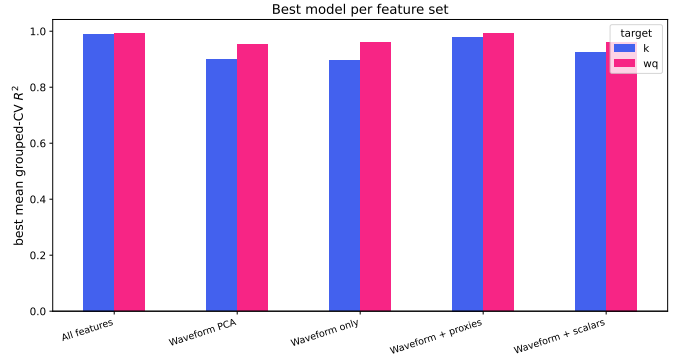


Fig. 1. Grouped-CV feature comparison using the best model for each feature-set/target pair. “Proxies” denotes synthetic geodesic features rather than physical shooting or ray tracing.

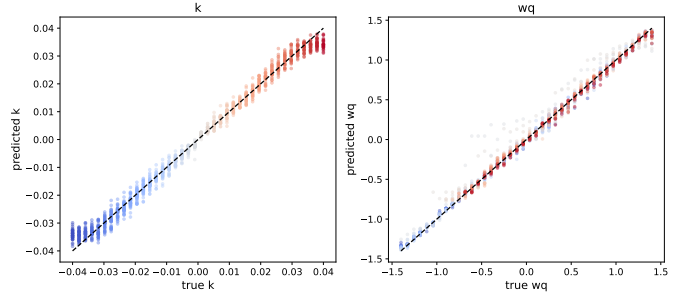


Fig. 2. True versus grouped-CV predicted k and w_q for the all-feature histogram-gradient-boosting model, including waveform PCA, scalars, and synthetic geodesic proxies.

simulator, additional parameter dependence improves inverse recovery. The physical analysis below tests whether validated observables retain that advantage; final inverse-model generalization is not inferred from the Jacobian alone.

The optional true-versus-predicted diagnostic in Fig. 2 shows the held-out structure for the best all-feature model. Residuals are not uniform over parameter space; this motivates the observability-classification task rather than treating a global score as a complete scientific result.

Waveform-only w_q MAE rises by more than 25% at the first tested nonzero noise level, 0.5%. This sensitivity should not be interpreted as a detector threshold: the perturbation is idealized Gaussian noise on normalized simulator curves and omits calibration, colored noise, mode uncertainty, start-time selection, and waveform systematics.

V. Learning Observability and Analytic Identifiability

Figure 3 compares the analytic class with predictions from the best all-feature random-forest classifier. Its macro F1 is 0.9928. The model captures both the broad transition in w_q and the narrow structure around small $|k|$. This is the central result: the classifier recovers where the inverse problem is informative and where finite-precision

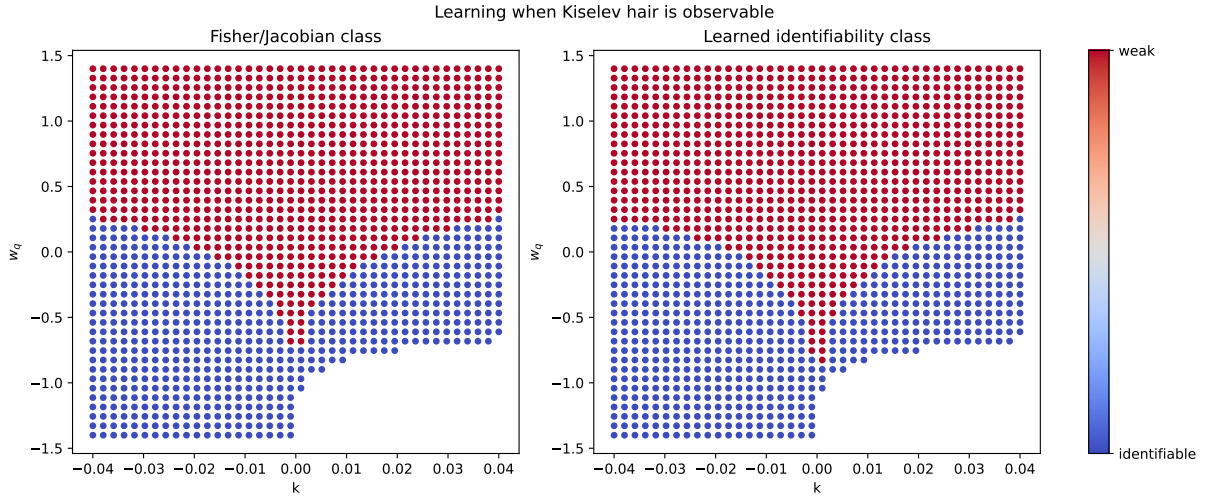


Fig. 3. True Fisher/Jacobian identifiability class (left) and grouped-CV prediction (right). Weak identifiability means $\sigma_{w_q} > 0.3$ under 1% relative observable precision. The all-feature random-forest classifier reaches macro F1 = 0.9928. The independently verified $k = 0$ limit is Schwarzschild; the nontrivial result shown here is the learned finite-precision boundary at small nonzero $|k|$.

uncertainty is strongly amplified, rather than merely returning a parameter estimate.

An independently evaluated $k = 0$ limit provides a structural sanity check: the metric reduces to Schwarzschild and w_q becomes unobservable. The nontrivial result shown by the sampled grid is the finite-width weak-identifiability region at small nonzero $|k|$.

A companion proxy-identifiability experiment provides a complementary view of this boundary in Fig. 4. Large condition number marks amplification of observable perturbations in the inverse problem, while the minimum singular value measures the weakest local parameter direction. In that companion scalar/proxy experiment, proxy augmentation reduces mean absolute w_q error from 0.0225 to 0.0143. In a near- $k = 0$ band, the median minimum singular value increases by approximately 12 $\times$, and the median condition number improves by approximately 2.13 $\times$. Improvement at small nonzero k does not remove the independently verified exact- $k = 0$ rank loss.

The agreement between learned classification and analytic sensitivity is more scientifically informative than the regression score alone. It distinguishes model failure from an inverse map whose local geometry suppresses information. At the same time, the comparison remains internal to the assumed formulas and error model. A different observable covariance or a validated perturbative spectrum would move the practical boundary.

VI. Validated Physical Geodesic Shooting

The proxy experiment motivated a separate physical calculation in the static Kiselev metric

$$ds^2 = -f(r)dt^2 + \frac{dr^2}{f(r)} + r^2 d\Omega_2^2, \quad f(r) = 1 - \frac{2M}{r} - \frac{k}{r^{1+3w_q}}. \quad (8)$$

An equatorial timelike emitter is evolved with conserved energy and angular momentum from apocentre. The saved coordinate azimuth begins at $\phi = \pi$, with $r = r_a$, $p_r = 0$, and emission time and proper time zero. Coordinate ϕ is not a Newtonian anomaly, so radial turning points are detected from $p_r = 0$ rather than imposed at multiples of 2π .

Photons use a three-dimensional Cartesian null Hamiltonian,

$$H_\gamma = \frac{1}{2} \left[-\frac{1}{f} + \mathbf{p}^2 + (f-1)(\mathbf{n} \cdot \mathbf{p})^2 \right] = 0, \quad (9)$$

and a two-angle root solve targets the static-observer plane. The direct branch exports redshift $1+z = \omega_{\text{emit}}/\omega_{\text{obs}}$, conserved impact parameter, signed observer-tetrad sky coordinates, propagation and excess delay, and direct arrival time. An independent curve uses

$$T_z(\phi) = \int_{\tau(\pi)}^{\tau(\phi)} (1+z) d\tau_{\text{emit}}, \quad (10)$$

with only the common additive constant removed. Failed phases are reported, not interpolated or replaced by proxy values.

For the Schwarzschild benchmark ($M = 1$, $r_p = 8M$, $r_a = 12M$, observer at $(0, 0, -80M)$), all 322 rays across two arbitrary w_q labels succeed and all observables agree exactly between labels. Maximum hit, timelike-constraint, null-constraint, and impact-drift errors are respectively 3.85×10^{-8} , 2.01×10^{-13} , 7.20×10^{-12} , and 1.78×10^{-11} . The arrival integral converges at approximately second order. Pericentre is at $\phi = 8.28375$ and the radial azimuthal period is 10.28431, giving physical apsidal advance 4.00113.

At fixed $w_q = -0.5$, changing k from zero to 10^{-3} produces resolved changes in every selected orbital, red-

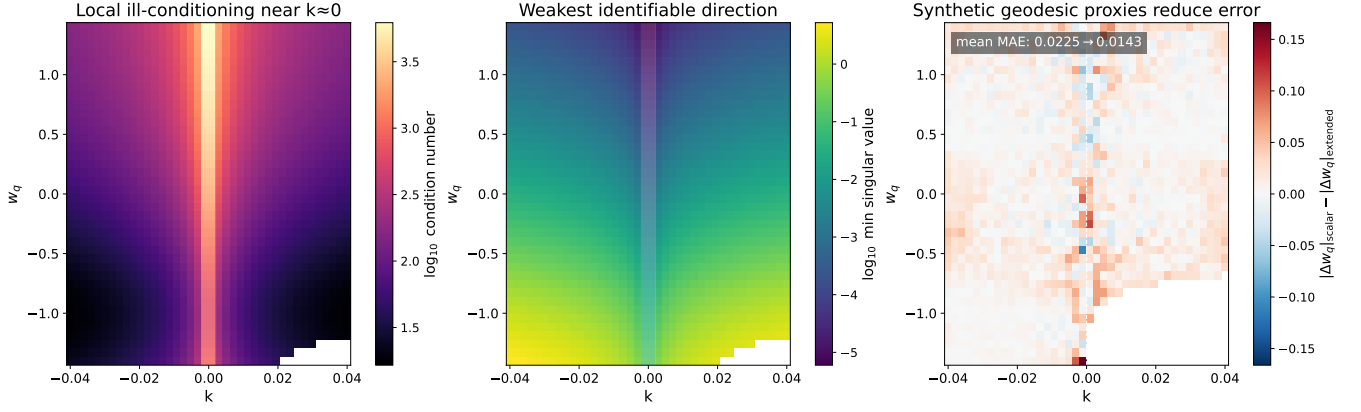


Fig. 4. Analytic identifiability lens. Left: Jacobian condition number. Center: minimum singular value. Right: reduction in w_q error after adding synthetic geodesic proxies. The proxies are not physical ray-tracing or shooting outputs.

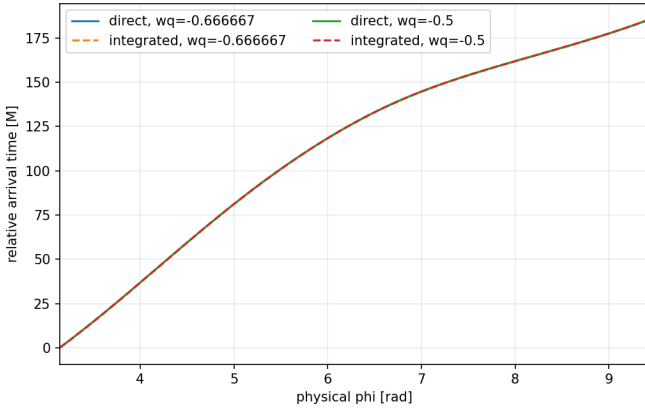


Fig. 5. Schwarzschild direct and integrated-redshift arrival curves. The trapezoidal residual is reported rather than forced to zero.

shift, photon-geometry, and timing observable; the apsidal advance changes by 0.12767. At fixed $k = 10^{-3}$, changing w_q from -0.5 to $-2/3$ changes the maximum redshift by 0.07341, impact parameter by 0.30504, excess delay by 3.2342, and relative arrival time by 1.6523. These ratios establish numerical resolvability, not observational signal significance.

VII. Physical Observable Complementarity Across Parameter Space

Local standardized sensitivity vectors show that large responses may still be nearly collinear. Timing has a large response but a sensitivity-vector angle of 178.76° and condition number 1275.5; photon geometry has the most distinct tested direction (159.37° , condition number 81.4). Adding more signals can therefore add redundancy rather than independent information.

A 9-by-9 admissibility scan identified 77 safe points, four invalid physical points, and no software failures. The selected 11-by-11 science grid contains 121 accepted

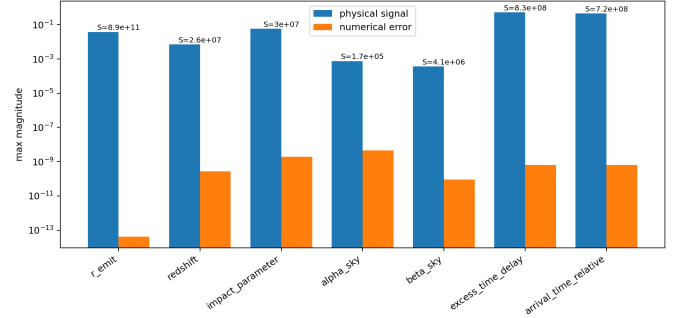


Fig. 6. First nonzero- k physical differences relative to numerical error.

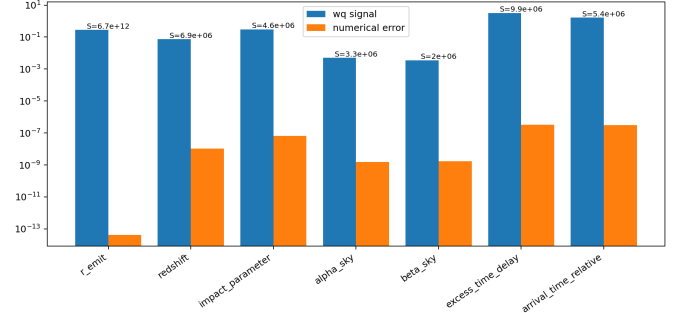


Fig. 7. Fixed- k physical sensitivity to changing w_q .

systems over $0 \leq k \leq 0.0025$ and $-0.7125 \leq w_q \leq -0.45$. At $k = 0$, the metric is independent of w_q and

$$\frac{\partial \mathbf{O}}{\partial w_q} = 0, \quad (11)$$

so exact rank loss is required physically. Twenty-seven 161-phase checks change the minimum singular value by at most 0.0401% and the condition number by at most 0.0713%.

Across the grid, ringdown plus all shooting has median minimum singular value 1.5312 and median condition

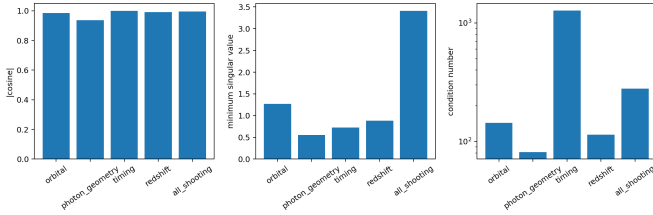


Fig. 8. Local sensitivity and conditioning by physical observable group.

TABLE II
Median normalized MAE across models, folds, and seeds.

Features	Random		Grouped		Directional	
	k	w_q	k	w_q	k	w_q
Ringdown	.0361	.1688	.0485	.3093	.0796	.3011
Photon geometry	.1088	.0927	.1813	.1111	.2291	.2224
All shooting	.0750	.0453	.1241	.0729	.2305	.2072
Ringdown+photon	.0378	.0730	.0457	.0874	.1305	.2012
Ringdown+all shooting	.0427	.0454	.0445	.0727	.1363	.2067

number 4.4974. Ringdown plus photon geometry gives 1.2326 and 4.0731, while all shooting gives 0.9723 and 6.0207. Relative to ringdown alone, ringdown plus photon geometry improves median minimum singular value by $4.54\times$ and median conditioning by $2.28\times$. Independent parameter direction, rather than feature count or signal amplitude, controls the gain.

VIII. Inverse Learning from Physical Shooting Features

We finally test whether the forward Jacobian predicts held-out inverse-model behavior. The 121-point table is evaluated with histogram gradient boosting, random forest, and a small target-scaled MLP over five deterministic seeds. Random interpolation, two contiguous 3-by-3 block layouts, and four directional extrapolation bands are kept separate. At $k = 0$, w_q is retained for boundary diagnostics but excluded from ordinary w_q scores because it is exactly unobservable.

Random splitting is optimistic: grouped/random degradation reaches 1.67 for k and 1.83 for identifiable w_q . Ringdown alone is strong for k but poor for w_q ; adding photon geometry reduces grouped w_q NMAE from 0.3093 to 0.0874. HGB and random forest both support this complementarity, while the MLP and individual extrapolation directions provide counterexamples to universal improvement. Aggregate directional ringdown+photon NMAE is 0.1305 for k and 0.2012 for w_q .

In a pooled explanatory model, standardized $\log \kappa$ has coefficient 0.197 with 95% bootstrap interval [0.185, 0.210], after including training distance (0.330 [0.317, 0.342]) and extrapolation status (0.276 [0.263, 0.287]); $R^2 = 0.144$. By contrast, the coefficient of $-\log \sigma_{\min}$ is consistent with zero. Conditioning remains informative, but coverage is stronger and no causal interpretation is claimed.

Split-conformal intervals target 90% coverage. Empirical k/w_q coverage is 89.2%/96.5% for random splits,

72.2%/85.1% for grouped regions, and 50.9%/66.0% for extrapolation. Formal exchangeability guarantees therefore do not extend to these shifted tests. Learning curves are still improving at 78 training points, and grouped one-percent feature noise gives ringdown+photon NMAE 0.0699/0.0853.

The principal failed check is phase-resolution robustness. On 27 independently recomputed points, replacing 81-phase shooting features with 161-phase values changes selected HGB predictions by up to 0.000614 in k , approximately 0.100 in w_q , and 0.093 in NMAE. The physical Jacobian is converged, but the inverse tree predictions are not uniformly robust to these small feature perturbations.

IX. Limitations and Future Work

Synthetic waveform scope. The inputs are generated by Eq. (3), not measured strain. Detector response, colored noise, calibration error, mode mixing, and ringdown start-time uncertainty are absent, so the noise experiment is a robustness test rather than an observational sensitivity forecast. The mode-averaged morphology is an equal-weight feature construction rather than an astrophysical multimode waveform with physically inferred amplitudes and phases.

Leading-eikonal/QNM scope. The geodesic-QNM correspondence is a leading-order approximation [1]. Dedicated perturbative spectra are needed to determine whether the same observability structure survives finite- ℓ corrections and waveform modeling error.

Proxy and physical-observable scope. The original workshop impact-parameter, screen-coordinate, delay, and redshift features remain synthetic proxies. They are preserved as a historical benchmark and must not be confused with the later Hamiltonian shooting outputs. Physical shooting is now implemented and numerically validated, but only for the direct branch, a point emitter and point observer, a static observer, compact dimensionless geometry, and without secondary images, radiative transfer, or a finite detector.

Validation and extrapolation scope. Grouped CV removes mode-level leakage but primarily measures interpolation across the sampled family. The physical feature grid is modest. Random, blocked-region, directional, conformal, and feature-noise tests are now reported, but learning curves remain data limited and shifted conformal coverage is below nominal. The 81/161-phase tree-prediction discrepancy must be resolved before final inverse-performance claims. Simulator-mismatch studies remain necessary.

The next ray-propagation step is secondary branches, radiative transfer, and a finite observing model. The next waveform step is validation against dedicated Kiselev perturbative calculations, including multiple modes and finite- ℓ corrections. The ringdown model remains leading-eikonal, and there is no strain-level likelihood. Only after the physical-grid inverse validation and these modeling

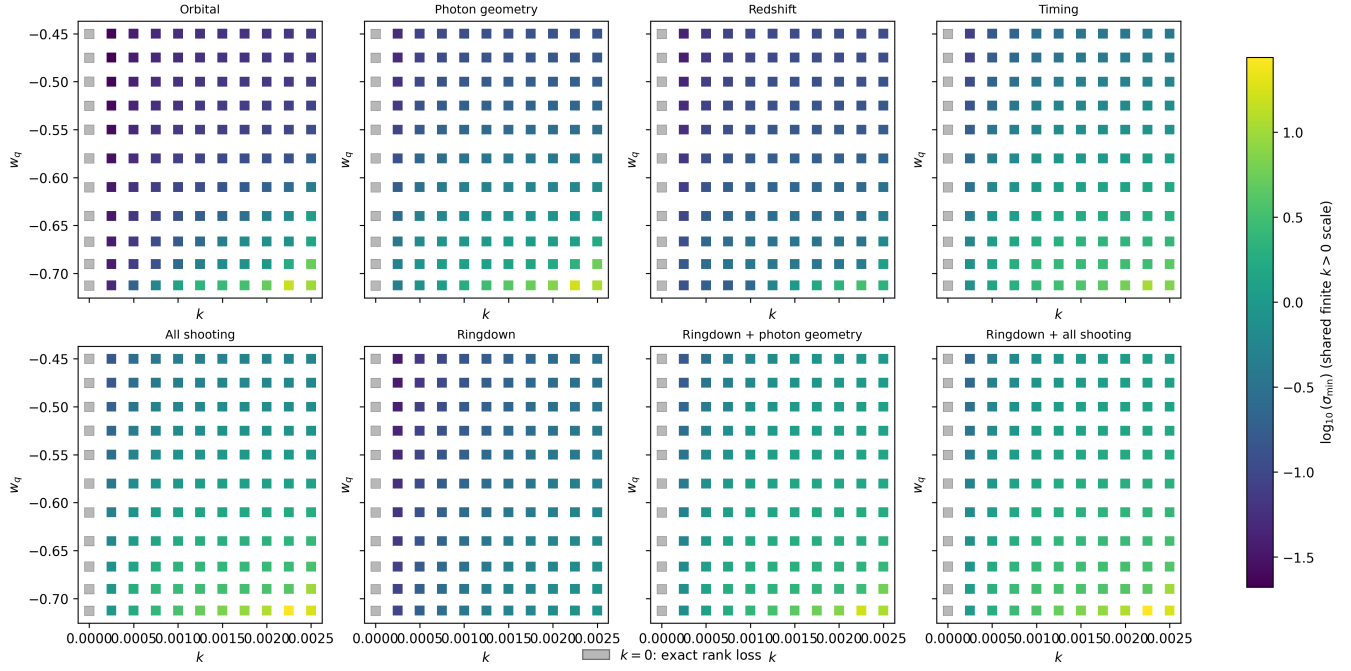


Fig. 9. Base-10 logarithm of the minimum singular value for each predeclared physical observable set. Exact $k = 0$ rank-loss points are masked in grey, and the shared colour scale is computed only from finite $k > 0$ values. Each square is a computed grid point; missing or invalid points are not interpolated.

checks should the framework be connected to public-event posteriors or a non-GR likelihood.

X. Conclusion

We presented a leakage-aware scientific-ML study of Kiselev waveform-to-hair inference. Grouping by physical parameters and fitting PCA only on training folds provide a credible interpolation test. Synthetic waveform morphology recovers (k, w_q) well inside the assumed simulator, while synthetic geodesic-proxy augmentation further improves regression. More importantly, an observability classifier learns a Fisher/Jacobian phase boundary with macro F1 0.9928, linking machine-learning errors to the local geometry of the inverse problem. The independently evaluated $k = 0$ limit establishes structural non-identifiability, while the sampled small-nonzero- $|k|$ region reveals full-rank but practically weak identifiability. The validated physical grid sharpens the interpretation: large timing or redshift signals may remain poorly identifiable when parameter responses are nearly collinear, whereas physically complementary ringdown and photon-geometry observables improve conditioning and grouped w_q inference in both tree families. Random validation is optimistic, directional extrapolation is harder, and conformal uncertainty does not fully recognize shift. The high-resolution substitution test prevents a claim of fully converged inverse performance. The broader AI-for-science principle is that inverse models should learn when parameters are observable—and expose when numerical or coverage limits invalidate a confident fit.

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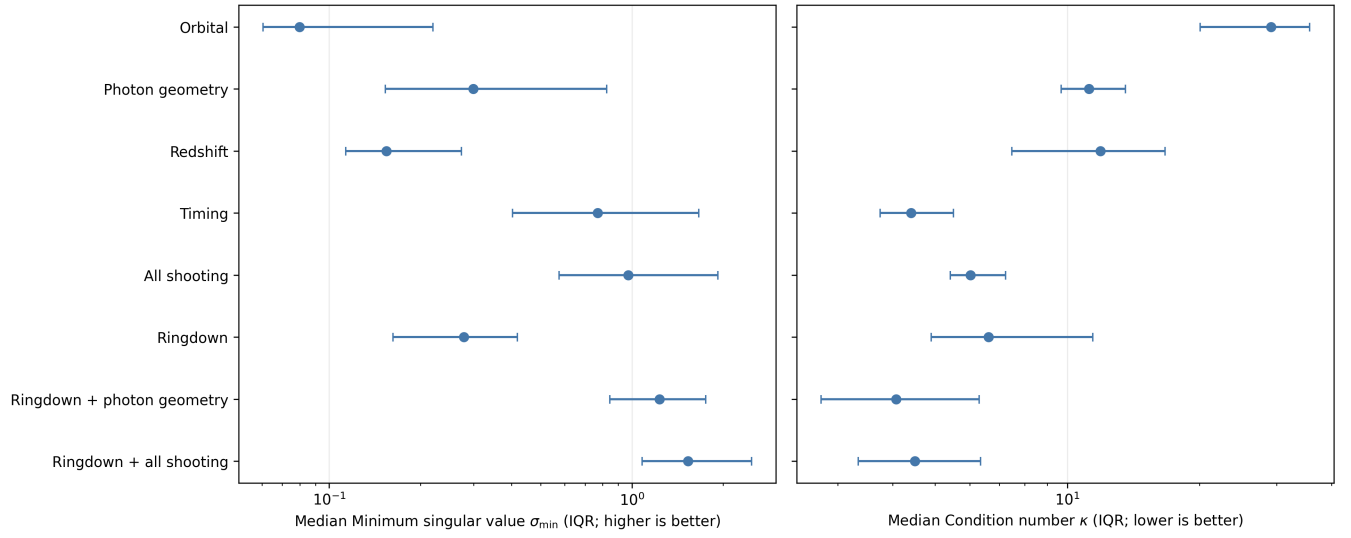


Fig. 10. Grid-wide median and interquartile range of the minimum singular value (left; higher is better) and condition number (right; lower is better), using finite $k > 0$ points. Full names identify every observable construction.

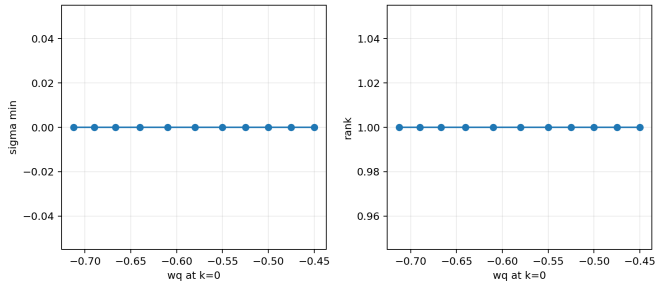


Fig. 11. Exact $k=0$ rank loss: w_q disappears from every physical observable.

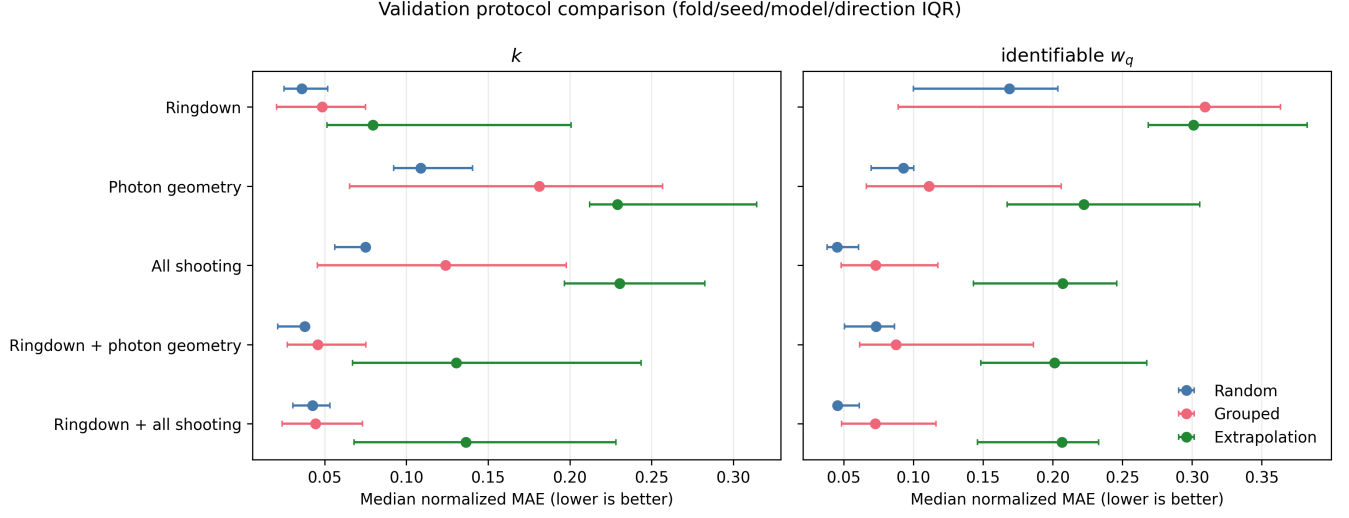


Fig. 12. Median normalized MAE for k (left) and identifiable w_q (right) under random interpolation, held-out contiguous regions, and directional extrapolation. Central values reproduce Table II exactly; intervals show fold/seed/model dispersion and, for the extrapolation summary, the four separately archived directions.

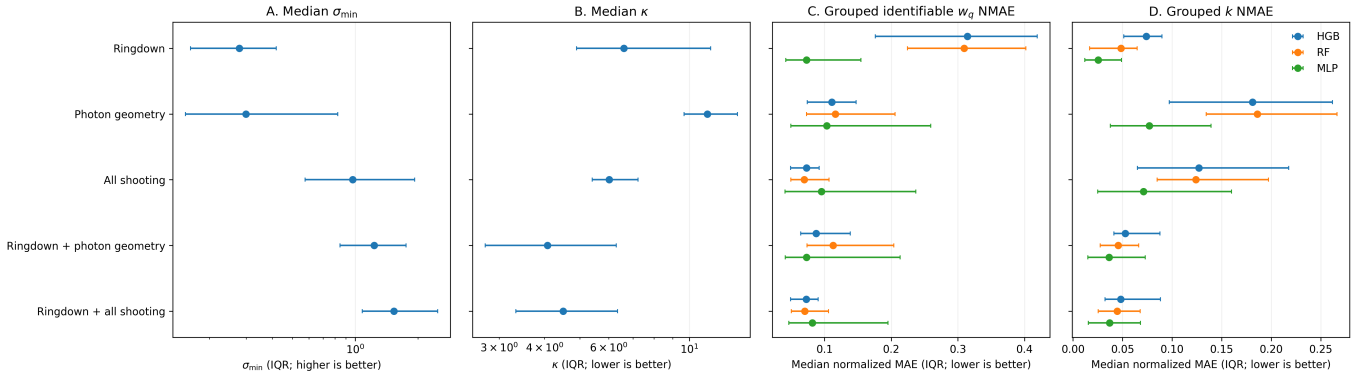


Fig. 13. Physical conditioning and grouped inverse-learning performance in a common feature-set order. Panels show (A) median minimum singular value (higher is better), (B) median condition number (lower is better), (C) grouped identifiable- w_q NMAE, and (D) grouped- k NMAE for HGB, RF, and MLP (lower is better). Physical conditioning generally agrees with tree-based grouped w_q performance, but does not perfectly rank every model, target, or protocol.

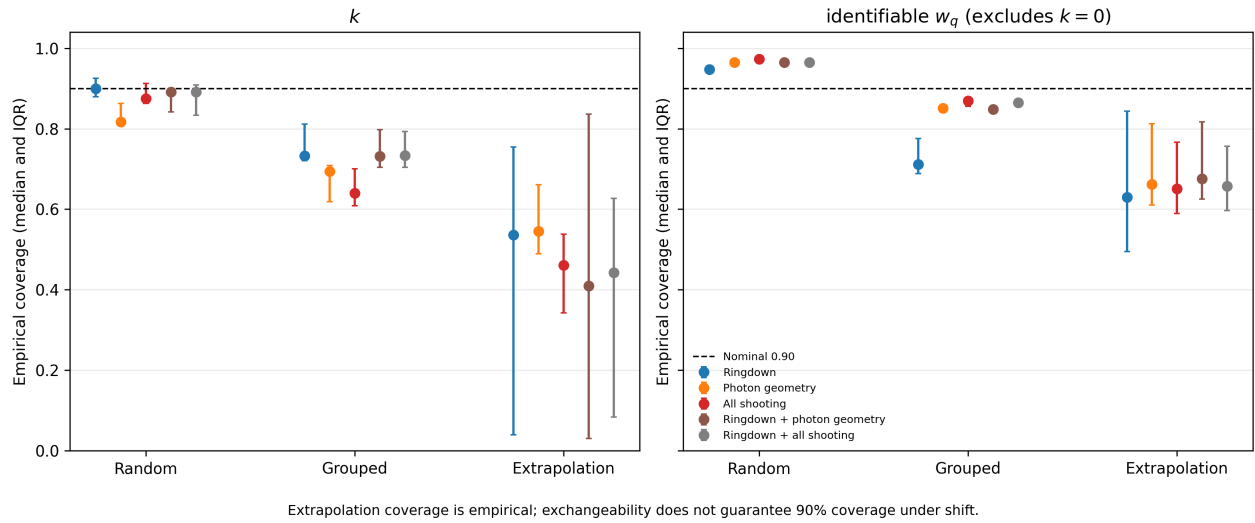


Fig. 14. Empirical split-conformal coverage for k and identifiable w_q (with $k = 0$ excluded from ordinary w_q coverage), separated by validation protocol. Points and intervals are the median and interquartile range across models (and directions where applicable); the dashed line is nominal 0.90 coverage. Extrapolation coverage is empirical and is not guaranteed under distribution shift.

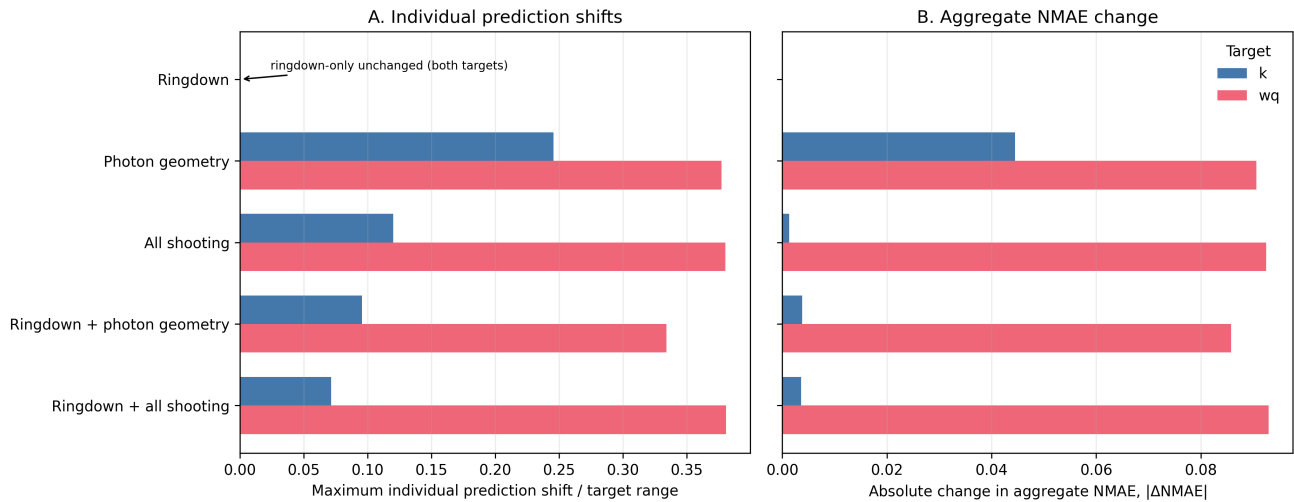


Fig. 15. High-resolution robustness when 161-phase shooting features replace 81-phase features at 27 validation points. Panel A reports the maximum individual prediction shift normalized by the target range; panel B reports the absolute change in aggregate NMAE. Targets are shown separately, and the ringdown-only construction is unchanged because it contains no shooting features.